2025-2026 SPRING SEMESTER

SE 216- SOFTWARE PROJECT MANAGEMENT

SOFTWARE DEVELOPMENT PLAN

SpotterAI

20240601048 - Mehmet Kahraman

20240601099 - Kerem Güneş

20220601058 - Utku Sertkaya

20230601068 - İsmail Yiğit Şendur

20230601003 - Onur Açar

20230601092 - Salikh İbragimov

~~20200601033 - Arda Eren Keleş~~

~~20230601031 - Burkay Gedik~~

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1.Overview

Problem Definition

Maintaining a healthy lifestyle has become increasingly difficult in modern society. Many individuals struggle to follow consistent exercise routines, maintain balanced nutrition, and manage their daily health habits. Although there are many fitness and diet applications available, most of them provide generic recommendations that are not fully personalized to individual users.

Additionally, many people rely on personal trainers or dieticians for guidance, which can often be expensive and not always accessible. Furthermore, some coaching services may rely on outdated practices or non-scientific advice. As a result, users may receive inconsistent guidance that does not adapt to their changing health conditions or lifestyle.

Another challenge is the lack of motivation and consistency among users. Many individuals start fitness programs but fail to maintain them due to lack of engagement, feedback, or accountability. Existing applications rarely combine personalized coaching, behavioral analysis, and motivational systems in a single platform.

Therefore, there is a need for a more intelligent and personalized system that can provide scientifically based recommendations, monitor user behavior, and motivate individuals to maintain healthier lifestyles in a sustainable way.

Background Information

Recent developments in artificial intelligence have enabled the creation of systems capable of analyzing large amounts of personal data and generating personalized recommendations. AI technologies are increasingly used in fields such as healthcare, fitness, and lifestyle management.

Modern fitness applications already use certain technologies such as step tracking, calorie counting, and basic workout planning. However, most of these systems still rely on static algorithms and do not continuously adapt to the user's behavior, health data, or lifestyle patterns.

The concept of AI-powered personal coaching is gaining attention because it can simulate the guidance of a personal trainer while being available at any time. By integrating user data such as activity levels, workout performance, nutrition habits, and daily routines, AI systems can generate more personalized recommendations.

Furthermore, behavioral analysis through artificial intelligence can identify patterns in user habits. This allows the system to predict when users are likely to skip workouts or lose motivation, and provide proactive suggestions to help them stay consistent.

Another important advancement is the integration of smart planning tools. For example, AI can create weekly meal plans and automatically generate grocery lists based on the required ingredients. These features help users manage their health-related tasks more efficiently.

These technological developments provide an opportunity to create a comprehensive AI-based system that supports individuals in maintaining healthier lifestyles.

2.High-Level Functionality

2.1 FUNCTIONAL REQUIREMENTS

- The system shall allow users to create an account and authenticate by logging in and logging out using an email address and password.

- The system shall display the daily AI-generated exercise plan and daily AI-generated meal plan on the home page after the user successfully logs in. The displayed plans shall include relevant details such as exercise names, sets, repetitions, intensity level, meal items, nutritional values, and visual illustrations where applicable.

- The system shall allow users to view detailed nutritional information (including calories and macronutrients) and allergen information (e.g., nuts, dairy, gluten) for each meal and each individual nutrient item where applicable..

- The system shall adjust the intensity of the exercise plan based on the user's sleep data, calorie intake data, and previous exercise performance, which are collected from a smartwatch and phone motion sensors.

- The system shall display users' profile pictures alongside their names and scores on the leaderboard and allow filtering and sorting by region, performance metrics (total points,completed workouts,workout streak), and time period (daily, weekly, monthly).

- The system shall generate a personalized workout plan using AI via the Hugging Face AI platform and shall display detailed visual illustrations for each exercise, including sets, repetitions, and proper form guidance.

- The system shall create a personalized diet plan based on the user's dietary preferences (preferred foods, dietary restrictions), nutritional targets (calorie intake, macros,micros), health goals (weight loss, fitness, muscle gain), and physical characteristics (weight, height, body fat percentage, age).

- The system shall use the Hugging Face AI platform to perform research-based analysis and generate personalized recommendations for meals (calorie intake, nutrient balance, dietary restrictions), supplements, and exercises (correct form, sets, repetitions, weight), using most recent peer-reviewed studies, user preferences, and user feedback.

- The system shall allow users to rate their previous meals and exercises at the end of each day using numerical ratings and optional text comments. The collected feedback shall be analyzed using Hugging Face AI to improve future personalized recommendations.

- Users who rank within the top 10% on the leaderboard shall be awarded rewards such as badges or supplement discounts, which shall be automatically assigned once per month after the leaderboard is updated and displayed on both the leaderboard and the user’s profile page.

- The system shall allow users to customize their profile by updating personal information (name, profile picture, bio) and to add, remove, or organize their favorite meals and exercises.

2.2 Non-Functional Requirements

- The system shall maintain 99% uptime availability.

- The system shall use the Scrypt algorithm for password hashing and AES-256 encryption for storing personal data (name, birth date, profile information), ensuring compliance with KVKK regulations

- The system shall ensure that users' meal plans and exercise programs are only visible to other users if the profile owner has explicitly allowed access, protecting sensitive dietary and exercise information.

- The app should work on both iOS (version 13.0 and above) and Android (version 8.0 and above).

- First time users should be able to sign in and login within 3 minutes, receive and apply exercise guidance within 5 minutes.

- Nutritional, allergen, and other displayed data shall be accurate and based on verified scientific sources to the best extent possible, and shall be regularly reviewed to ensure consistency and reliability.

- The system shall use OAuth 2.0 for authentication in login and registration processes.

- User sessions shall automatically expire after 15 minutes of inactivity.

- The system shall communicate via RESTful APIs.

- AI analysis of feedback shall be completed within 5 seconds to update recommendations.

3.Software Tools

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Development of the AI Personalized Recommendation & Optimization Engine
2	Scalable Backend Infrastructure & Health Data Security Management
3	Development Environment Setup for AI and Backend Implementation

SOFTWARE TOOLS FOR TASK 1: Development of the AI Personalized Recommendation & Optimization Engine

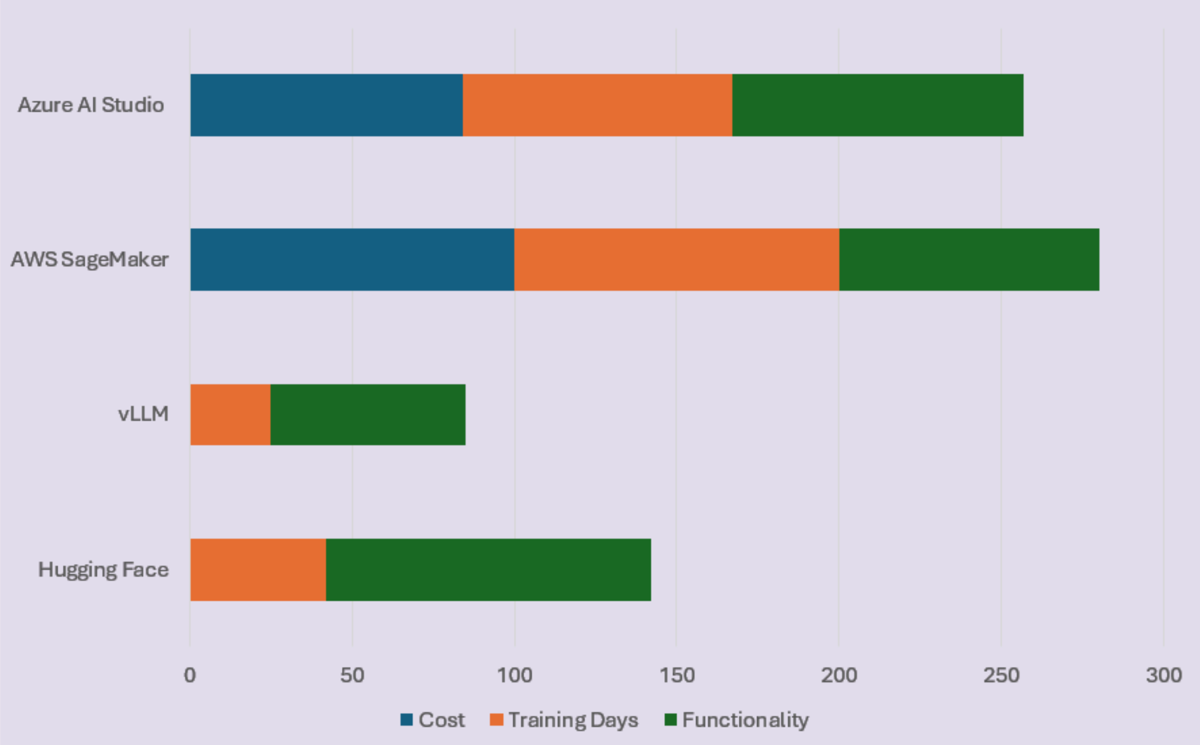
Tool Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	$0 (Open Source)	$0 (Open Source)	$500/mo	$420/mo
Training Days	5	3	12	10
Functionality	10	6	8	9

Normalized Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	0	0	**100**	84
Training Days	42	25	**100**	83
Functionality	**100**	60	80	90

Normalized Tool Graph



Which tool has been selected? Why?

Hugging Face was selected.

Why: Getting pre-trained models up and running quickly was our top priority here, alongside painless fine-tuning. Hugging Face ended up being the obvious pick. Rather than spending almost two weeks wrestling with massive cloud environments like SageMaker just to deploy a tweaked model, their ecosystem lets us pull pre-trained weights, fine-tune them on our custom dataset, and get everything live in a fraction of the time. Plus, keeping the base cost at zero while still getting top-tier usability gives the team way more breathing room when we inevitably need to scale up the infrastructure later.

SOFTWARE TOOLS FOR TASK 2: Scalable Backend Infrastructure & Health Data Security Management

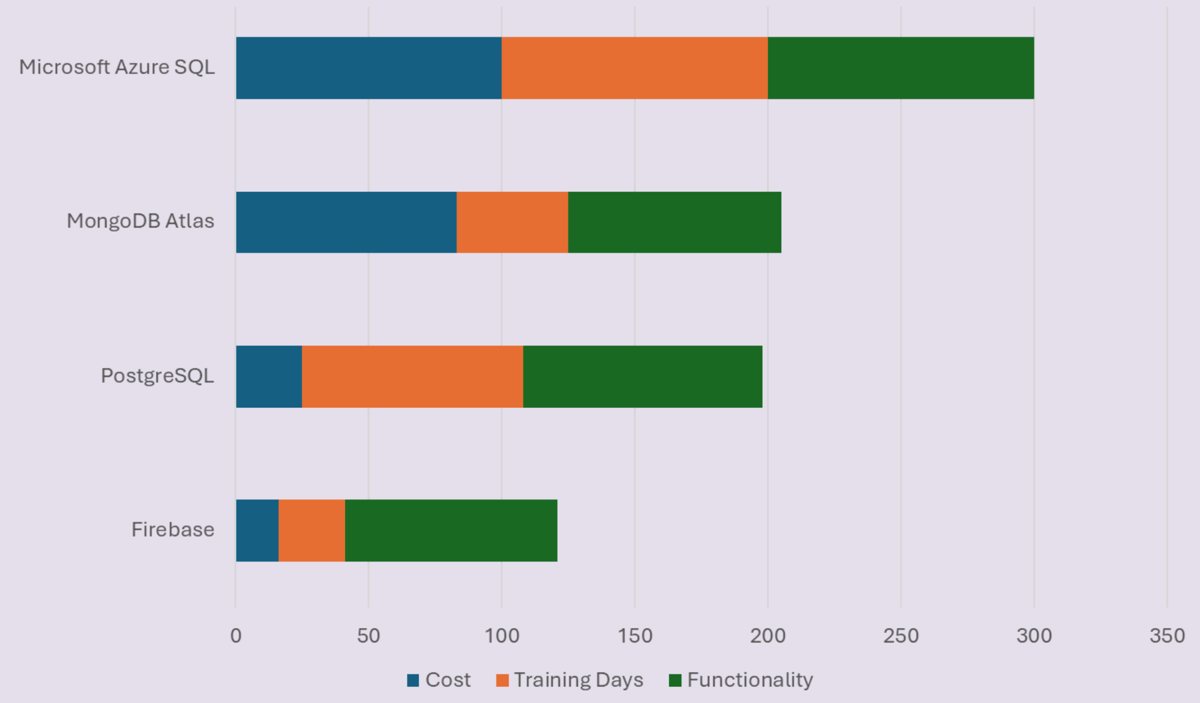
Tool Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	20/mo	30/mo	100/mo	120/mo
Training Days	3	10	5	12
Functionality	8	9	8	10

Normalized Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	16	25	83	**100**
Training Days	25	83	42	**100**
Functionality	80	90	80	**100**

Normalized Tool Graph



Which tool has been selected? Why?

PostgreSQL was selected.

Why: PostgreSQL was selected for its strong ability to manage large-scale structured data, support complex queries, and ensure high data integrity. It guarantees that database operations are processed reliably and consistently, meaning that data is not lost, partially updated, or left in an inconsistent state even in case of system failures. This level of reliability is essential for handling sensitive health and fitness data. Additionally, PostgreSQL provides advanced features such as joins, indexing, and high scalability, making it a robust and efficient solution for backend development.

SOFTWARE TOOLS FOR TASK 3: Development Environment Setup for AI and Backend Implementation

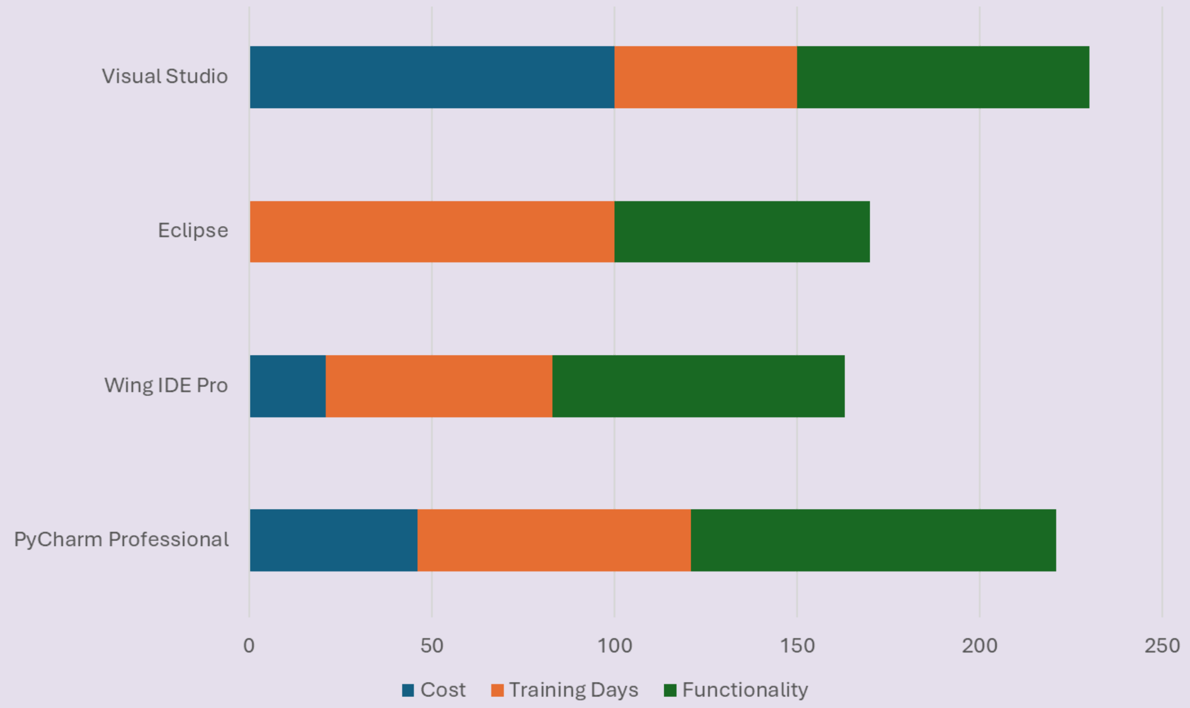
Tool Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	$1500/yearly	$690	$0	$3240/yearly
Training Days	6	5	8	4
Functionality	10	8	7	8

Normalized Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	46	21	0	**100**
Training Days	75	62	**100**	50
Functionality	**100**	80	70	80

Normalized Tool Graph



Which tool has been selected? Why?

PyCharm Professional was selected as the development environment for Python because it provides advanced features such as intelligent code completion, powerful debugging tools, integrated testing support, database management, and seamless integration with AI and backend frameworks. Although it is a paid solution, its high functionality and enterprise-level capabilities make it the most suitable IDE for large-scale and AI-based projects.

4. Software Process Model

We chose the Scrum Model for our project because it is flexible, fast, and perfect for team collaboration. Our application "AI Personal Health & Fitness Coach" has complex features like AI integration and medical warnings, which require continuous testing and adaptation.

4.1 Product Backlog

#	Product BackLog
1	Database planning and optimization for meals, exercises, and user data
2	Implementation of Scrypt password hashing and AES-256 encryption for user data
3	Implementation of backup and disaster recovery system
4	Development of user registration (email & password) and login/logout system
5	User profile management (name, profile picture, bio editing)
6	Implementation of premium membership system (advanced AI features, plans)
7	Integration of in-app purchase functionality
8	Setup of AI infrastructure and selection of the most suitable AI model based on project objectives

9	Customization and fine-tuning of the AI model using our custom datasets
10	Automation of scientific paper analysis to customize the AI model, and integration of RESTful API to connect the AI system with research data
11	Integration of Hugging Face API for personalized workout plan generation
12	Development of an AI-based meal plan and exercise plan generation system
13	Development of exercise plan components including sets, repetitions, intensity, and visual guidance
14	Design interactive home screen layouts for daily AI-generated meal and workout plans, optimizing readability and user engagement
15	Development of push notification system (exercise reminders, meal alerts)
16	Development of leaderboard and ranking algorithm
17	Creation of challenge system (weekly goals and competitions)
18	Scientific studies analyses and AI-generated summaries
19	Integration with Apple Health and Google Fit
20	Development of admin panel for user and content management

4.2 Sprints

Sprint-1 (8 days)

- Database planning and optimization
- User registration system
- Premium membership system

Sprint-2 (8 days)

- In-app purchase functionality
- User profile management
- Development of admin panel

Sprint-3 (15 days)

- AI Infrastructure
- Hugging Face API Integration
- Scientific AI-generated summaries
- Customization and fine-tuning of the AI model

Sprint-4 (9 days)

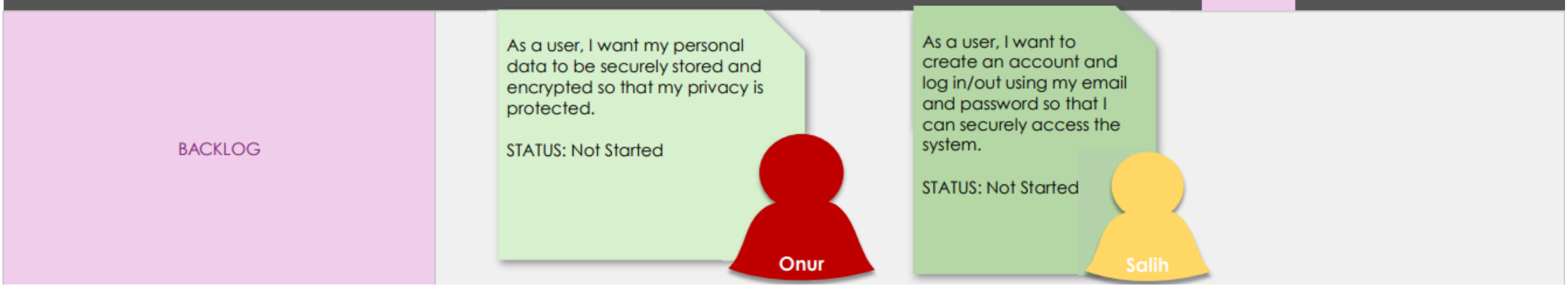
- AI-based meal plan and exercise plan generation system
- Development of exercise and meal plan components
- Design interactive home screen layouts
- Development of push notification system

Sprint-5 (9 days)

- Integration with Apple Health and Google Fit
- Creation of challenge system
- Leaderboard and ranking algorithm

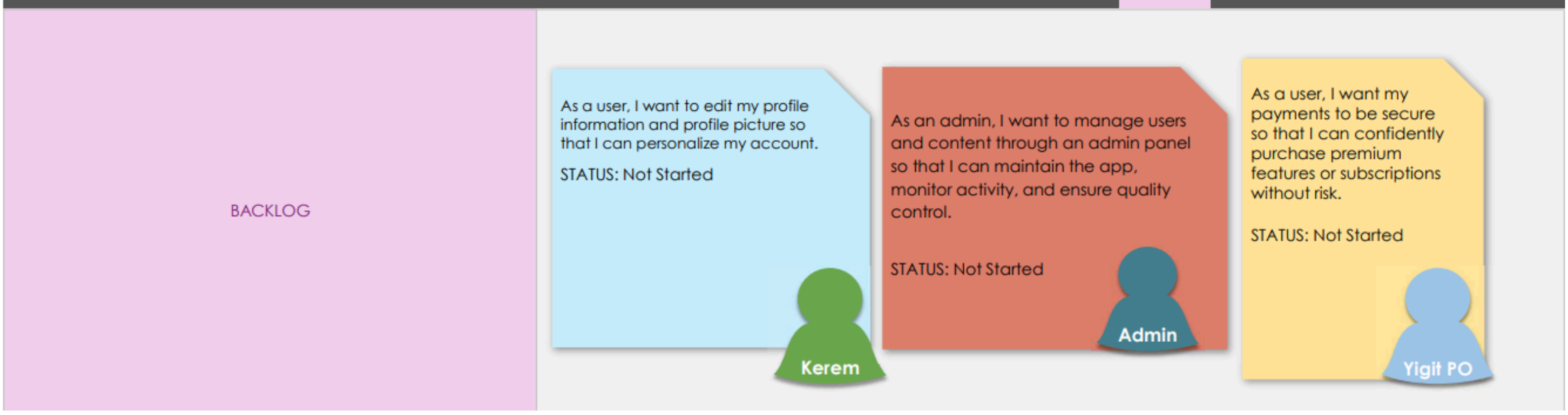
SPRINT-1

STAFF	ACTION	STATUS	PRIORITY	START	FINISH	DURATION	NOTES AND COMMENTS
Developer, Product Owner	Database planning and optimization	Not Started	High	16.02.2026	18.02.2026	4	Ensure compatibility with future AI-based meal/exercise plan integration
Developer	User registration system	Not Started	High	19.02.2026	23.02.2026	3	Add optional 2FA to increase account security
Developer, Product Owner	Premium membership system	Not Started	Medium	19.02.2026	23.02.2026	2	Define premium tiers and features clearly
						8	



SPRINT-2

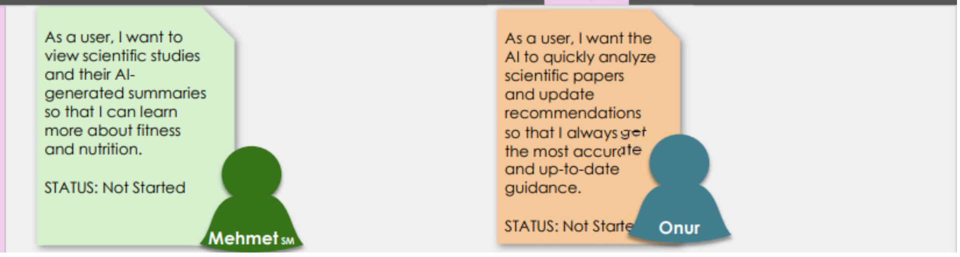
STAFF	ACTION	STATUS	PRIORITY	START	FINISH	DURATION	NOTES AND COMMENTS
Developer	In-app purchase functionality	Not Started	Medium	24.02.2026	26.02.2026	3	Integrate with Apple Store and Google Play billing systems
Developer	User profile management	Not Started	Medium	27.02.2026	3.03.2026	4	Ensure changes are reflected immediately in app UI
Developer, Product Owner	Development of admin panel	Not Started	Medium	2.03.2026	3.03.2026	3	Follow data protection regulations (KVKK) and best practices
						8	



SPRINT-3

STAFF	ACTION	STATUS	PRIORITY	START	FINISH	DURATION	NOTES AND COMMENTS
Developer	AI infrastructure	Not Started	High	4.03.2026	5.03.2026	2	Ensure scalability and future integration with meal/exercise plan systems
Developer	Hugging Face API Integration	Not Started	High	6.03.2026	10.03.2026	3	Ensure secure API keys and rate limiting
Developer	Scientific AI-generated summaries	Not Started	High	11.03.2026	16.03.2026	3	Ensure AI-generated summaries match original research findings
Developer	Customization and fine-tuning of the AI model	Not Started	Medium	13.03.2026	18.03.2026	3	Report Training metrics (loss, accuracy)
						15	

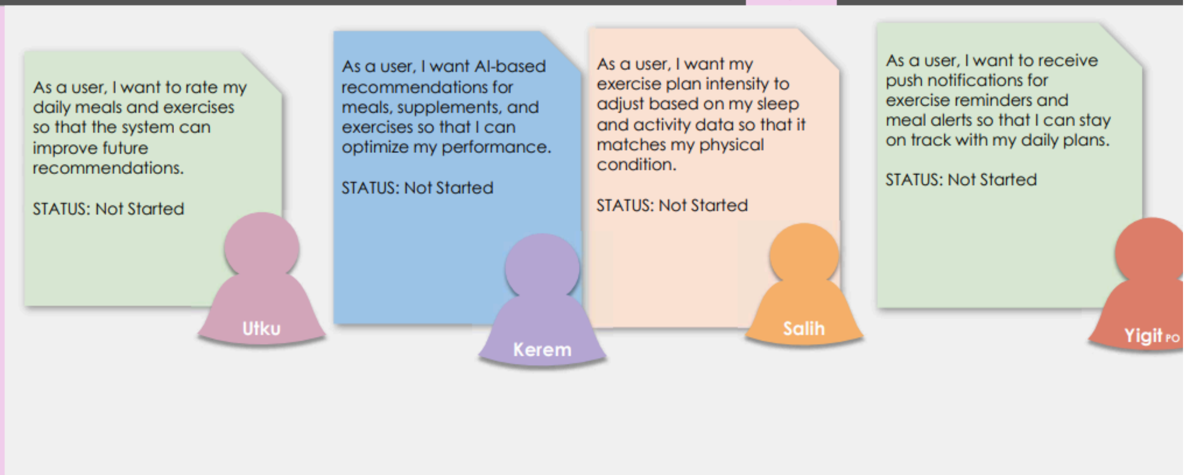
BACKLOG



SPRINT-4

STAFF	ACTION	STATUS	PRIORITY	START	FINISH	DURATION	NOTES AND COMMENTS
Developer	AI-based meal plan and exercise plan generation system	Not Started	Medium	19.03.2026	23.03.2026	3	Use user data and preferences to personalize recommendations
Developer	Development of exercise and meal plan components	Not Started	Low	24.03.2026	26.03.2026	3	Check that the user sees the same plan that the AI created
Developer	Design interactive home screen layouts	Not Started	Low	24.03.2026	27.03.2026	3	Test with different screen sizes and devices
Developer	Development of push notification system	Not Started	Low	27.03.2026	27.03.2026	1	Allow users to customize notification times
						9	

BACKLOG



SPRINT-5

STAFF	ACTION	STATUS	PRIORITY	START	FINISH	DURATION	NOTES AND COMMENTS
Developer	**Integration with Apple Health and Google Fit**	**Not Started**	**High**	**30.03.2026**	**1.04.2026**	**3**	**Ensure compatibility with multiple devices**
Developer	**Creation of challenge system**	**Not Started**	**Medium**	**2.04.2026**	**7.04.2026**	**4**	**Notify users about challenge start/end**
Developer, Product Owner	**Leaderboard and ranking algorithm**	**Not Started**	**Medium**	**3.04.2026**	**7.04.2026**	**3**	**Handle edge cases appropriately**
						9	

BACKLOG

As a user, I want the app to integrate with Apple Health and Google Fit so that my fitness and health data from other devices are automatically synchronized with the app.
STATUS: Not Started

Salih

As a user, I want to filter and sort the leaderboard so that I can analyze rankings easily.
STATUS: Not Started

Utku

As a user, I want to participate in weekly challenges and competitions so that I stay motivated and engaged with my exercise and meal plans.
STATUS: Not Started

Onur

5. Schedule and Effort

5.1 Project Schedule

ALL	TASK NAME	DURATION	PLANNED STA...	PLANNED FINIS...	LINKED FRO...
1	**Sprint 1**	**6 days**	**16.02.2026**	**23.02.2026**	
2	Database planning and optimization	3 days	16.02.2026	18.02.2026	
3	User registration system	3 days	19.02.2026	23.02.2026	
4	Premium membership system	3 days	19.02.2026	23.02.2026	
5	**Sprint 2**	**5,5 days**	**24.02.2026**	**3.03.2026**	
6	In-app purchase functionality	3 days	24.02.2026	26.02.2026	
7	User profile management	2,5 days	27.02.2026	3.03.2026	
8	Development of admin panel	1,5 days	2.03.2026	3.03.2026	
9	**Sprint 3**	**11 days**	**4.03.2026**	**18.03.2026**	
10	AI infrastructure	2 days	4.03.2026	5.03.2026	
11	Hugging Face API Integration	3 days	6.03.2026	10.03.2026	10
12	Scientific AI-generated summaries	4 days	11.03.2026	16.03.2026	11
13	Customization and fine-tuning of the AI model	4 days	13.03.2026	18.03.2026	
14	**Sprint 4**	**7 days**	**19.03.2026**	**27.03.2026**	
15	AI-based meal plan and exercise plan generation system	3 days	19.03.2026	23.03.2026	
16	Development of exercise and meal plan components	3 days	24.03.2026	26.03.2026	15
17	Design interactive home screen layouts	4 days	24.03.2026	27.03.2026	
18	Development of push notification system	1 day	27.03.2026	27.03.2026	
19	**Sprint 5**	**7 days**	**30.03.2026**	**7.04.2026**	
20	Integration with Apple Health and Google Fit	3 days	30.03.2026	1.04.2026	
21	Creation of challenge system	4 days	2.04.2026	7.04.2026	20
22	Leaderboard and ranking algorithm	3 days	3.04.2026	7.04.2026	

5.2 Gantt Chart

6. Stakeholders and Project Risks

This analysis is focusing on stakeholders and project risks.

6.1 Stakeholders

#	STAKEHOLDER	DESCRIPTION
1	**Users**	If the project is successful, users benefit by improving their health, saving time, and receiving useful guidance. If it fails, they may get incorrect advice, waste time, and lose trust in the application.
2	**Project Owner**	If the project is successful, the owner gains profit, recognition, and business growth. If it fails, they may lose money and face damage to their reputation.
3	**Health & Fitness Experts**	These are people who give health and fitness knowledge to improve the app. If the project is successful, they help make better and safer recommendations. If it fails, wrong advice may reduce trust in them.
4	**Fitness Centers / Gyms**	These are places where people do exercise and fitness activities. If the project is successful, they may get more customers and business deals. If it fails, they get no benefit.
5	**Food & Nutrition Companies**	These companies provide healthy food products and meal ideas. If the project is successful, they may get more business and partnerships. If it fails, they get no advantage.
6	**Service Providers**	Service providers such as cloud platforms, AI APIs, and hosting services support the technical infrastructure of the project. If the project is successful, they benefit from increased usage and revenue. If it fails, they may lose a client and potential income.
7	**Wearable Technology Companies**	These companies make devices like smartwatches and fitness trackers. If the project is successful, they can connect their devices with the app and get more users. If it fails, they lose this opportunity.
8	**Universities and Research Centers**	These are sources of scientific research and papers used to improve the project. If the project is successful, their studies help us build better and more accurate features. If it fails, we may not benefit effectively from their research or use it correctly.

9	**Obesity Treatment Centers**	Medical centers that help manage weight and obesity. If successful, they can support patients with better tracking and recommendations. If it fails, there is no added benefit or collaboration.
10	**Competitor Companies**	These are other companies with similar apps. If the project is successful, they have more competition and try to improve. If it fails, nothing changes for them.

6.2 Project Risks

LIKELIHOOD RANK	RISK DESCRIPTION
1	**Changing Requirements :**Project requirements may change during development. This can cause confusion and extra work.
2	**No Real Data :** To test "Habit Prediction," we need to know how people behave. Since the app isn't out yet, you don't have real data to see if our code actually works.
3	**Integration Issues :** Different parts of the system may not work well together. This can slow down development.
4	**Schedule delays :** AI and backend development may take longer than expected.
5	**Low accuracy of AI model :** Generated workout/nutrition plans may be incorrect or ineffective.
6	**Testing :** Evaluating the accuracy of AI-generated nutrition and workout plans is difficult without a large set of real-world health data or a controlled testing environment.
7	**API Costs :** Running out of money for AI tokens during dev.
8	**Dependency :** One person leaving the team or getting sick.
9	**Stakeholder Misalignment :** Project stakeholders (team, advisor, users, etc.) may have different expectations and goals. This can lead to confusion, repeated changes in requirements, and delays in development.

10	**KVKK / GDPR / Health Authority Compliance Risk** : Improper handling of health/user data may violate privacy laws and health authority requirements, leading to legal issues or inability to deploy the app

IMPACT RANK	RISK DESCRIPTION
1	**Low accuracy of AI model :** Generated workout/nutrition plans may be incorrect or ineffective.
2	**No Real Data:** To test "Habit Prediction," you need to know how people behave. Since the app isn't out yet, you don't have real data to see if our code actually works.
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4	**KVKK / GDPR / Health Authority Compliance Risk** : Improper handling of health/user data may violate privacy laws and health authority requirements, leading to legal issues or inability to deploy the app .
5	**Stakeholder Misalignment :** Project stakeholders (team, advisor, users, etc.) may have different expectations and goals. This can lead to confusion, repeated changes in requirements, and delays in development.
6	**API Costs:** Running out of money for AI tokens during dev.
7	**Schedule delays:** AI and backend development may take longer than expected.
8	**Testing** : Evaluating the accuracy of AI-generated nutrition and workout plans is difficult without a large set of real-world health data or a controlled testing environment.

9	**Changing Requirements:** Project requirements may change during development. This can cause confusion and extra work.
10	**Dependency:** One person leaving the team or getting sick.

LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
2	2	4	**No Real Data:** To test "Habit Prediction," you need to know how people behave. Since the app isn't out yet, you don't have real data to see if our code actually works.
5	1	6	**Low accuracy of AI model :** Generated workout/nutrition plans may be incorrect or ineffective.
3	3	6	**Integration Issues:** Different parts of the system (AI, backend, database) may not work well together. This can slow down development.
1	9	10	**Changing Requirements:** Project requirements may change during development. This can cause confusion and extra work.
4	7	11	**Schedule delays:** AI and backend development may take longer than expected.
7	6	13	**API Costs:** Running out of money for AI tokens during dev.

6	**8**	**14**	**Testing:** Evaluating the accuracy of AI-generated nutrition and workout plans is difficult without a large set of real-world health data or a controlled testing environment.
10	**4**	**14**	**KVKK / GDPR compliance risk:** Improper handling of user data may cause legal issues.
9	**5**	**14**	**Stakeholder Misalignment :** Project stakeholders (team, advisor, users, etc.) may have different expectations and goals. This can lead to confusion, repeated changes in requirements, and delays in development.
8	**10**	**18**	**Dependency:** One person leaving the team or getting sick.

7. Measurements

7.1 Identifying Measurements

Questions to identify measurements:
-Is the project staying within the API and infrastructure budget? **-How much effort went into gathering, cleaning, and labeling the health and fitness data?** **-How accurate is the AI in predicting user habits and behavior patterns?** **-What is the total volume of health data and scientific research successfully processed for the AI model?**
Identified measurements:
• **Monthly API cost / Server cost /Estimated budget** • **Weekly staff-hours /Data size** • **Correct predictions / Total predictions** • **Number of successfully processed data rows/Number of incorrectly labeled data rows / Total MB of cleaned data**

Measurement storage and collection:

What: API/Cloud infrastructure costs and staff-hours spent on project activities.

- **When:** Weekly (for team effort) and monthly (for budget billing cycles).
- **Format:** Real numbers (Currency/USD) and Integers (Hours).
- **How:** Total **active hours** are collected directly from the **Scrum board** during daily updates. The Project Manager matches these actual hours with the monthly **cloud provider invoice** and records them in the master spreadsheet.

What: Number of correct/incorrect rows, successfully processed rows, and total MB of cleaned data.

- **When:** After each data cleaning sprint or main database update.
- **Format:** Integer counts (for rows) and Real numbers (for MB).
- **How:** Extracted via **MySQL queries** and validation scripts, then entered into the project documentation by the Data Engineering team.

What: Correct AI predictions vs. total predictions.

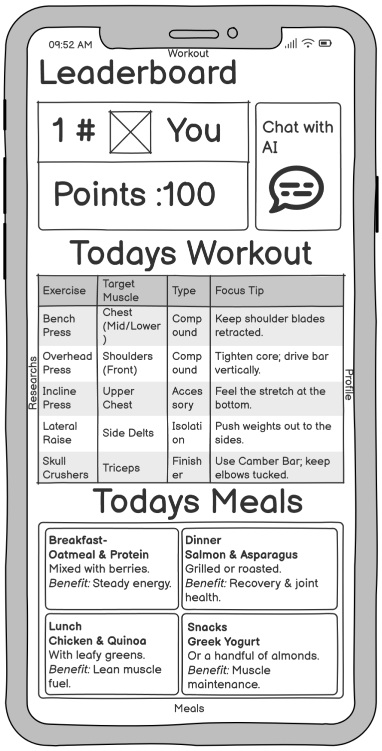
- **When:** After each model training iteration or automated testing session.
- **Format:** Integer counts (raw predictions) and Percentages (accuracy rate).
- **How:** Automatically calculated by the testing scripts and logged directly into the AI performance tracking dashboard.

7.2 Measurement Types

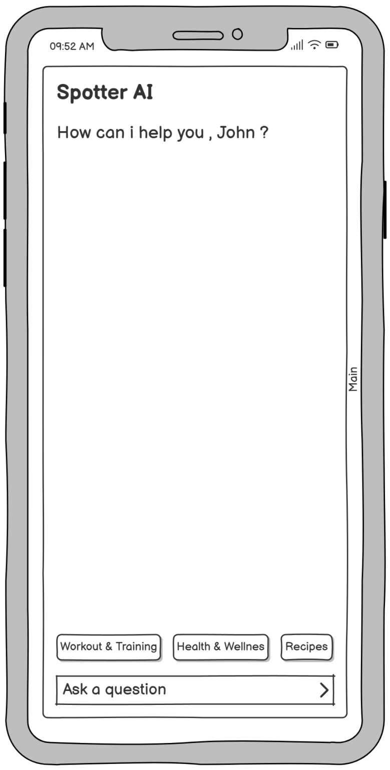
Measurement Type	Description	Example Measurements
Budget	Compares actual project costs against the planned budget	$250 spent / $400 estimated in Month 2.
Team Effort	Measures team productivity by tracking the total amount of work completed through **Story Points**	21 Story Points completed in Sprint 1.
AI Accuracy	Evaluates if the AI model provides logical and accurate fitness and nutrition recommendations to the user	48 correct and helpful meal/workout plans / 50 total generations = 96% accuracy.
Data Quality	Measures the sensors data (GPS, steps, gyroscope) per Megabyte, factoring in the precision and variety of the tracking hardware	12,000 rows with full sensor variables (GPS + Accelerometer) / 50 MB = 240 valid rows per MB.
Data Processing Efficiency	Measures how effectively the system collects, stores, and processes large amounts of project data.	8,500 successfully processed rows/10,000 collected data rows = **85% efficiency**
Defect Density	Measures the percentage of collected data that is accurate, error-free, and directly usable for analysis	9,200 accurate rows / 10,000 total collected rows = 92% usability rate.

8.Graphical User Interfaces

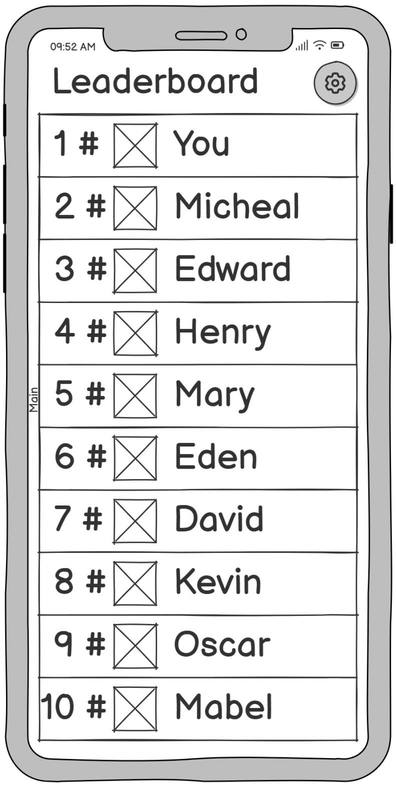
This part demonstrates the use of interfaces and capable features of the project.

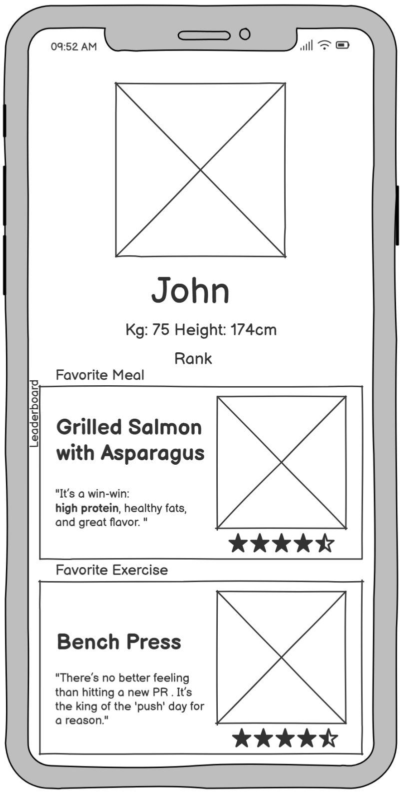


This dashboard shows your daily progress, leaderboard, workout routine, and meal plan in one place.

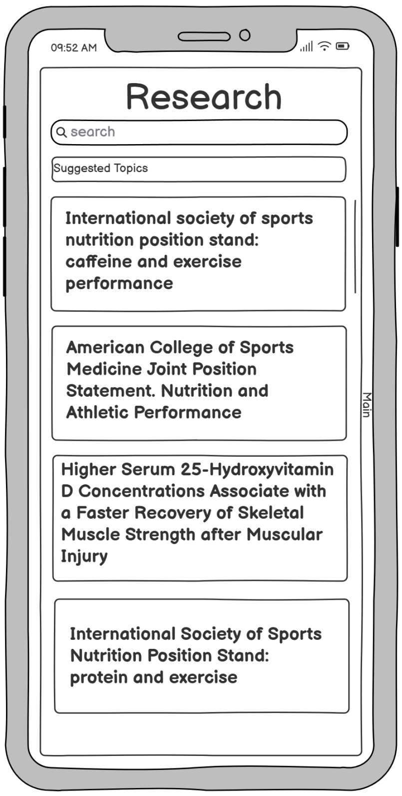


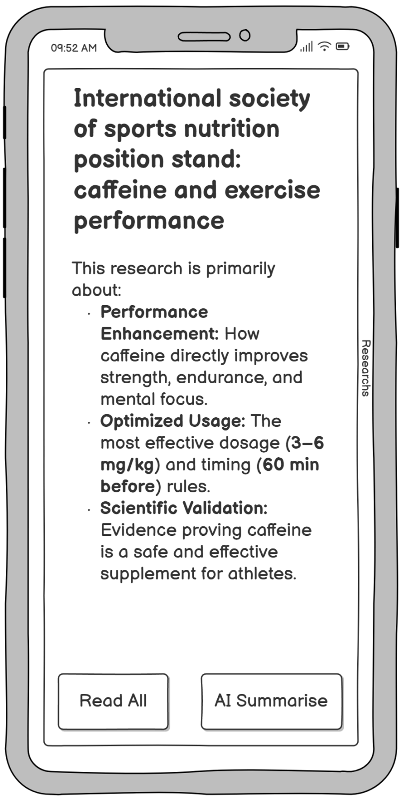
This screen lets you chat with an AI assistant to get quick answers about workouts, food, and health.



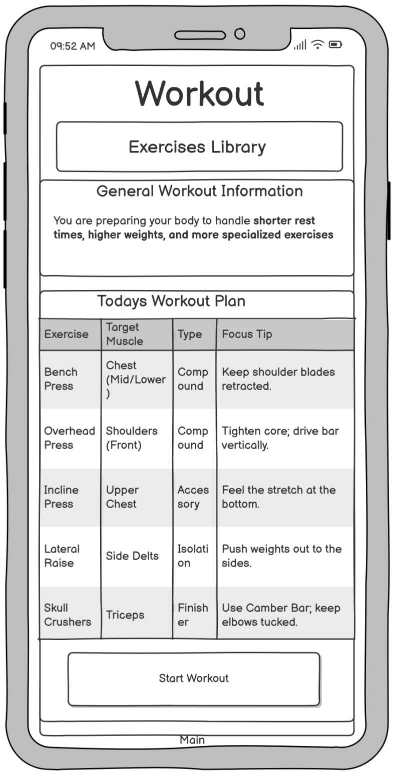


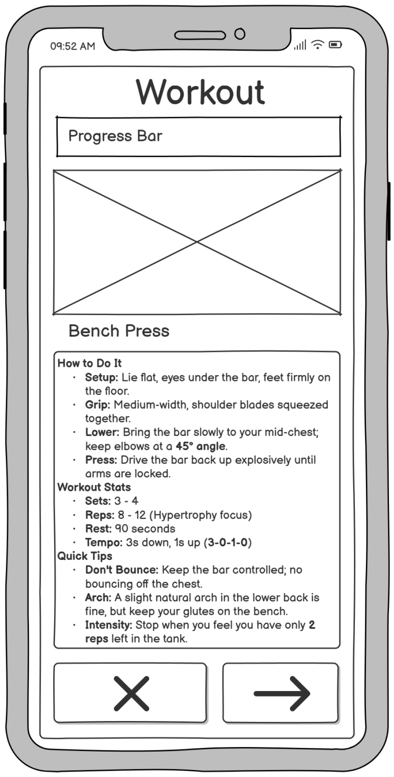
The leaderboard shows user rankings, letting you click on profiles to see their favorite exercises and meals.

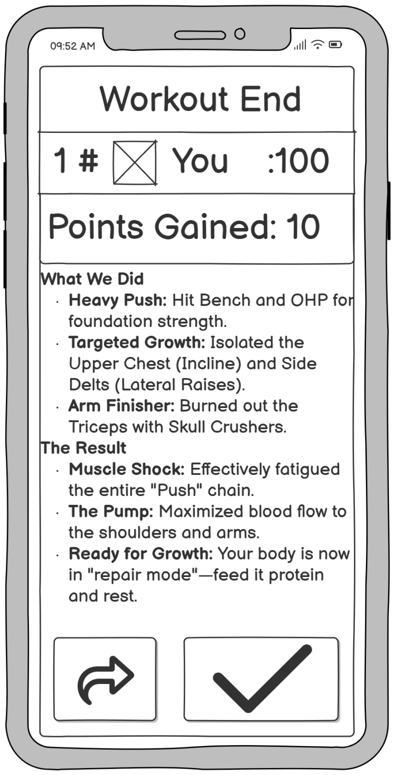


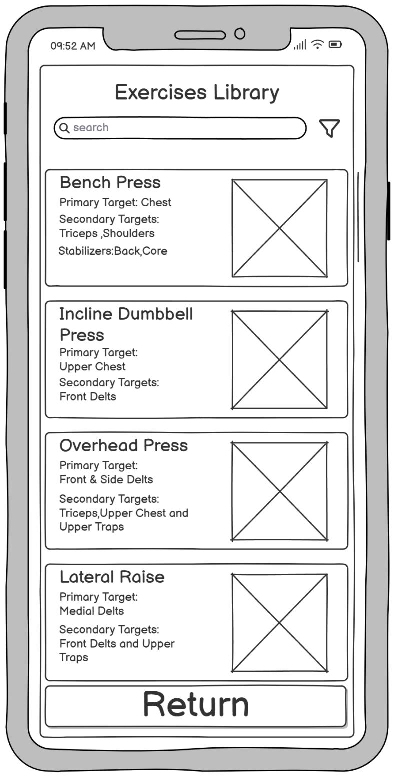


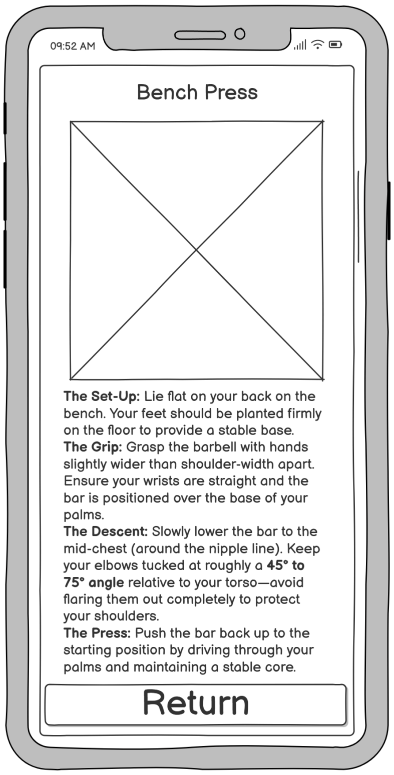
The Research tab lets you browse scientific articles to read general information and get quick AI-generated summaries.



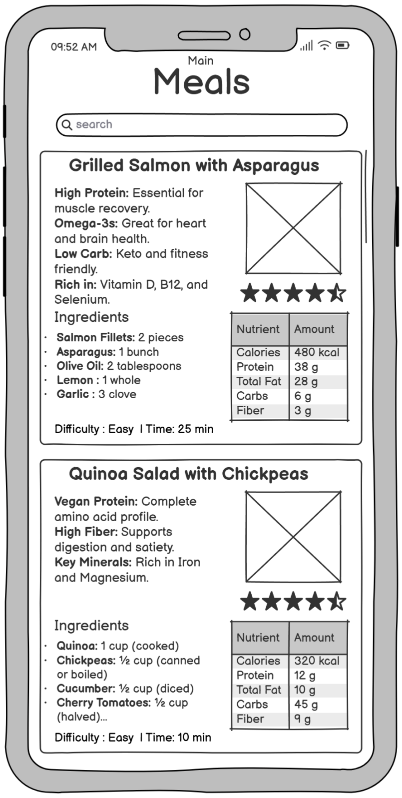


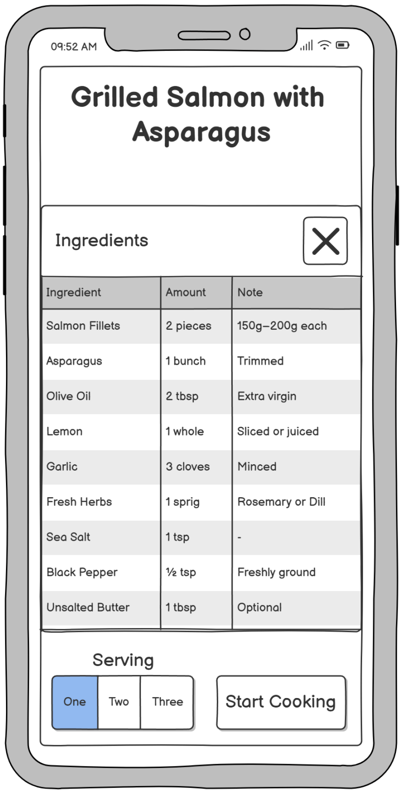


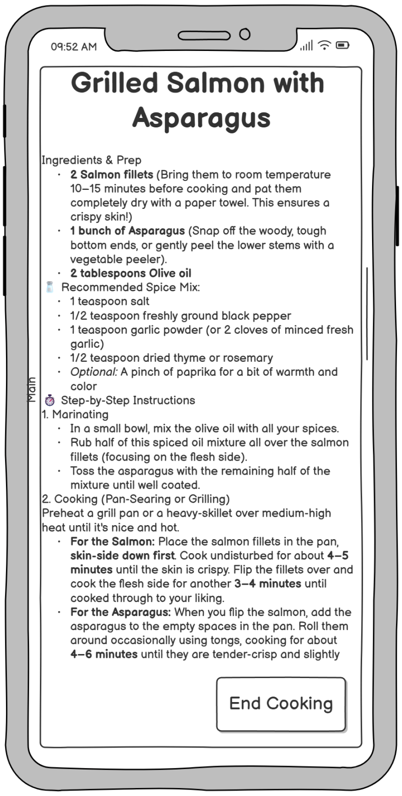


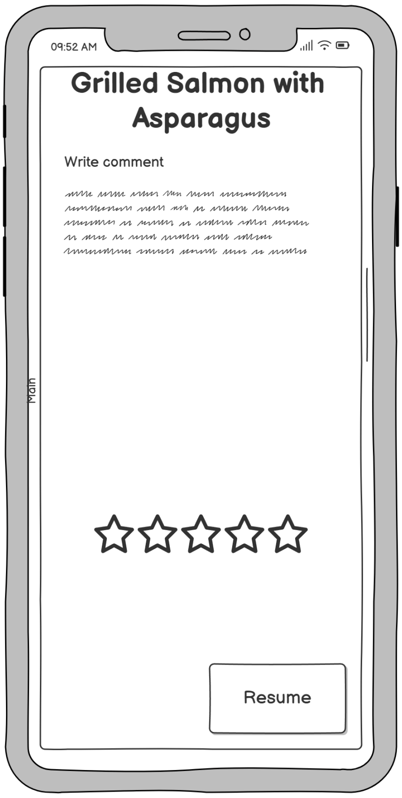


The Workout tab allows you to start training sessions to see your workout results at the end, while the Exercise Library provides access to detailed guides on how to perform each movement.









In the Meals tab, you can view meals, nutrition facts, and ingredient lists, and you can also rate and write your comments at the end of the meal.

The Profile tab lets you view and edit your profile to rate and review your favorite exercises and meals, while the settings and notifications screens let you manage your app preferences.

9. Conclusion

In conclusion, this project was developed to create an AI-supported fitness and nutrition application called SpotterAI for users who want to improve their health habits. The main goal of the project was to provide personal exercise and meal suggestions based on the user's goals and progress. The application also includes features such as progress tracking and leaderboard systems to increase user motivation.

During the project, we worked on different parts of the software development process such as requirement analysis, project planning, risk identification, and GUI design.

Although the current version of the project is still a prototype, it shows how artificial intelligence can be used in fitness and health applications. In the future, the system can be improved with more advanced AI features, wearable device support, and better recommendation systems.

By integrating artificial intelligence with everyday health management, SpotterAI has the potential to become a reliable digital fitness partner helping users build sustainable habits, stay motivated, and ultimately lead healthier lives.